



MIT SLOAN SCHOOL OF MANAGEMENT
MIT SCHWARZMAN COLLEGE OF COMPUTING



MODULE 2 UNIT 4
Podcast 1 Transcript

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Five candidates, one choice

PAUL MCDONAGH-SMITH: A week later, the same six people came back to the same room. The whiteboard hadn't been wiped. Rebecca's two underlined phrase was still there. At the top right corner, a little faded but unmistakable. Ambitious but unprepared.

The homework from the first meeting had been specific. Each member of the executive team was to bring to this second Friday one candidate, one workflow process, or function that they believed was the right first serious agentic deployment for Boulton Industries. Not the safest choice. Not the most ambitious. The one that would teach the most.

They had taken the instruction seriously. Five candidates landed on the table. The CEO had decided in the week between the meetings that she wouldn't bring one of her own. She wanted, she said, to adjudicate rather than advocate.

I want to walk you through the five candidates because the debate about which one to pick, which took the first 90 minutes of the meeting, is where the team began to learn the shape of the decisions that come with building agentic systems.

Vikram, the Chief Technology Officer, had brought an internal research assistant, an agent that would sit across Boulton's accumulated knowledge base, decades of engineering reports, project closeouts, technical studies, and help staff find, compare, and synthesize what the company already knew. Low stakes, high value, and Vikram argued the natural wave two progression into wave three.

Olu, the General Counsel, had brought contract review triage, an agent that would read inbound contracts, flag deviations from Boulton's standard terms, and then route them to the right reviewer. A clear efficiency win in a function under severe pressure. Governance built in from the start because the agent's output was advisory, not binding.

And then Rebecca, the Chief Financial Officer, had brought supplier qualification and onboarding. Boulton processes roughly 4,000 supplier applications per year, each one requiring financial health checks, sanctions screening, ESG disclosures, technical fit assessment, and a commercial decision. The end-to-end average to undertake this process is 36 days. Her proposal: an agent that could run the qualification process end-to-end with human sign-off at defined thresholds.

Maria, the Chief Operating Officer, had brought a shift handover summarization agent for the Shrewsbury manufacturing facility. Every shift change involves a handover from the departing shift supervisor to the arriving one. Production status, issues, open actions. Maria's proposal: an agent that sat in on the handover, produced the written summary, tracked open actions across shifts, and surfaced recurring patterns.

And then Priya, the Chief People Officer, had brought a new joiner onboarding companion, an agent that would walk a new hire through Boulton Industries systems, answer their questions in the first 30 days, and track which parts of the onboarding process they had and hadn't completed. Her argument: the domain where Boulton had the most human time

locked in repetitive context setting and the best opportunity to learn agent management skills at a low-stakes interface. And so, five candidates all defensible and all different.

The CEO opened the discussion with a single sentence: “Which of these teaches us the most?” And I watched the team spend the next hour working through that question. And I want to tell you where they landed and why because the reasoning illustrates I think something true about the first real agentic decision most organizations have to make.

Bounded on purpose

MCDONAGH-SMITH: The team ruled candidates in and out not by the usual criteria, not by which would generate the fastest ROI or which was lowest risk or which had the most visible executive sponsor. They ruled candidates in and out by asking which one would surface the most decisions about how Boulton Industries should build agentic systems.

Vikram’s research assistant was impressive, Olu noted. But it was mostly retrieval. The agent wouldn’t take consequential actions in the world. It would not need deep thresholds for human oversight. It would be genuinely useful and genuinely bounded. But it wouldn’t force Boulton Industries to confront most of the questions that agentic deployment potentially raised.

And then Priya’s onboarding companion had a similar shape. Low stakes, essentially advisory. Excellent for learning agent management as a soft skill but light on the architectural and governance decisions the team needed to practice making.

Maria’s shift handover agent was closer. It would touch real operational data and the summaries would genuinely influence decisions on the factory floor. But the risk surface was narrow and the handover context was specific enough that it wouldn’t teach Boulton much about cross-functional agentic workflows.

Olu’s contract triage was a strong, strong candidate. It involved judgment under uncertainty, real documents, meaningful downside if the agent got things wrong. Olu herself made the case against it: “It’s close to what I do for a living. I want us to pick the one where the team isn’t already the experts. We’ll learn more if we’re uncomfortable.”

That left Rebecca’s supplier qualification proposal, multistep, cross-functional. It touched procurement, finance, compliance, and operations. It involved real money, real suppliers, real risk, and crucially, had a baseline the team could measure against.

Thirty-six days, 4,000 applications a year, a known cost structure. You could tell precisely whether the agent was doing better than the current process, the same as the current process or worse. “This one will force us to make decisions we’d rather not make,” Olu said. Which is the point. If we pick the easy one, we won’t learn how to make the hard ones.

The CEO agreed. The supplier qualification agent became the selected candidate. And here, just as the team was about to move into architecture, Vikram said something that I want to slow down and give to you in full. Because it would frame what followed. He said, “Before we design this thing, we need to be honest about what kind of problem we’re solving. We’re not building an agent that will perform supplier qualification the way a senior procurement manager would. We are not trying to replicate human judgment. We are trying

to build an agent that will handle the bounded, well-defined, data-rich parts of supplier qualification so that the senior procurement manager spends her time on the parts that are genuinely ambiguous, bounded on purpose.”

And he let that phrase sit in the room for a second: “Bounded on purpose.” Then he told the team he had been reading *The Economist*, Herbert Simon, the mid-20th century researcher who won the Nobel Prize for his work on bounded rationality. “Simon’s insight,” Vikram said, “was that real-world decision-makers, including the very best ones, never necessarily solve the whole problem. They solve a simplified version of it inside a frame they have drawn deliberately using the information they can get to. And the quality of the decision depends on the quality of the bounding.”

“Agents,” Vikram said, “are the same. We don’t get anywhere by building agents that try to do everything. We get somewhere by building agents whose boundaries are drawn well. What can the agent see? What actions is it allowed to take? What does it remember? What does it hand off? The boundaries are the architecture.” And he walked to the whiteboard and he drew five boxes.

The five building blocks

MCDONAGH-SMITH: Vikram’s five boxes were labeled from left to right: perception, action, planning, memory, safety. He said these were the five decisions every agentic system forces you to make and that the supplier qualification agent would force Boulton Industries to make each of them consciously.

The first box, perception. What does the agent see? “For supplier qualification,” Vikram said, “the agent would need access to Boulton’s ERP records, to public financial databases, to sanctions lists, to ISO certification registries, and to the internal record of past decisions about similar suppliers.” Every one of those data sources was a deliberate choice. Each one had security implications. Each one had to be connected. And Boulton’s data, as Priya had observed in the first meeting, didn’t yet flow cleanly between its own systems.

The second box, action. What is the agent allowed to do? And Vikram listed the candidate actions. Query a database. Retrieve a document. Populate a scoring form. Draft an email to a supplier. Schedule a follow-up. Escalate to a human reviewer. “Notice,” he said, “that none of the actions I’ve listed commit Boulton Industries to anything. No contract signed. No purchases placed. No supplier approved without a human signing off. The agent proposes and the human approves. That’s an architectural decision, not a policy decision. It’s built into what the agent can and cannot do.”

And the third box, planning. How does the agent decide what to do next? And here Vikram made a distinction that mattered for the rest of the meeting. A traditional automation system follows a predefined workflow: Step 1, Step 2, Step 3. An agentic system decides the steps itself based on what it perceives. That flexibility is what makes it useful on complex variable problems. It’s also what makes it hard to predict. The agent will not always do the same thing on the same output. It will do the most sensible thing given the input it has. Which means our testing regime can’t be the kind we use for rule-based software. We’ll need a different approach.

And the fourth box, memory. What does the agent remember? Vikram distinguished two kinds of memory. The first was conversational. What had happened inside a single supplier qualification from beginning to end? The second was institutional. What the agent learned from previous qualifications that should help inform the next one. “And that second kind,” he said, “was where most organizations got it wrong because it was where bias and error propagation compounded silently.”

He said, “If we build an agent that remembers every decision Boulton Industries has ever made about a supplier, we’re also building an agent that remembers every mistake Boulton has ever made. And we need to decide deliberately which parts of our institutional memory are worth inheriting.”

And so, to the fifth box, safety. How do we prevent the agent from doing something harmful? Whether it’s by its own error or because someone tried to manipulate it. Vikram’s framing here was the one that Olu picked up on most quickly. Don’t rely on the agent itself to manage risk. Rely on the system around the agent. The agent’s outputs go through checks. Its actions have limits. Its escalations are verified by code, not by language. The safety logic lives outside the model because the model exactly the wrong place to put it. That sentence, “Don’t rely on the agent to manage risk,” became, in the hours that followed, one of the governing principles of the Boulton Industries design.

The five boxes stayed on the whiteboard. Vikram said the team would return to each of them before the meeting ended. And that, he said, was the architecture the team would need to design together.

Build, buy, or stack, and how fast?

MCDONAGH-SMITH: And so, Olu was the first to push the conversation into the next register. “Architecture was one question,” she said, “Procurement was another.” Are we building this ourselves or buying a platform or assembling it from components?

Vikram had expected this question. He had a three-column sketch already drawn in his notebook. He put it up on the screen. “Building it yourselves,” he said, “meant hiring or contracting the engineering capability, choosing the foundational models, integrating with Boulton’s data sources, and owning the operational responsibility.” Maximum control, maximum learning, maximum effort. The longest time to first deployment. Buying a platform meant licensing a vendor’s pre-built agent infrastructure and configuring it to Boulton’s first use case. Fastest time to first deployment. Least engineering overhead. But also least control. The vendor decides what the underlying architecture is. How the agent reasons. Where the data lives, and what happens when the vendor changes their roadmap.

Stacking meant choosing components from multiple providers. A foundation model from one, an orchestration layer from another, a monitoring tool from a third, and integrating them yourselves. A middle path. More control than buying, faster than building, but it introduced what Olu immediately named as the integration risk. The risk that the stack you assemble today would stop working the moment one of its components evolved.

The debate that followed was arguably the sharpest of the meeting. Rebecca made a case for buying. She was not, she said, in the business of funding capability-building experiments. She wanted a working system in production within the quarter with a vendor who took operational responsibility for the parts Boulton didn’t understand.

Vikram made a case for building. The whole point of the exercise, he said, was to develop Boulton Industries' capability to design agentic systems. Buying one from a vendor would deliver a working agent without developing the organizational muscle to design the following next 10 agents, for example.

Maria made a case for stacking but with a warning. She had watched the manufacturing industry work through a similar argument a decade earlier over industrial control systems. She said, "Whichever way you go, one rule applies. Don't rely on any one single vendor. Build the architecture so you can switch components without rebuilding the whole system. The technology will change faster than your procurement cycle."

And so, the CEO adjudicated. The decision, she said, would be a stack with Boulton doing enough of the engineering to learn, enough of the buying to move at the pace the board expected, and enough deliberate decoupling between components that no single vendor decision would lock the organization in. She said, "We're not optimizing for speed of first deployment. We're optimizing for rate of learning at an acceptable speed."

And then Maria raised the other question I'd been waiting for. "Three weeks from design to minimum viable products," she said, "that's not enough time to think." The proposal on the table, implicit in Rebecca's original candidate, was a minimum viable product within three weeks, followed by a measured evaluation period, followed by a decision on whether to expand. Maria's instinct, honed over 20 years in manufacturing, was that three weeks was too fast. We don't deploy anything in manufacturing in three weeks. We test, we iterate, we commission.

But Vikram pushed back, carefully. The three-week MVP, he said, was not a production system. It was a learning artifact. The point was not to scale it. The point was to build it fast enough to find out what you didn't know and then to evaluate it deliberately before doing anything else. He said the alternative is a six-month evaluation cycle. And in this technology, a six-month evaluation cycle is itself a form of risk. By the time you finish evaluating, the thing you evaluated isn't the thing in front of you anymore.

Olu agreed. However, with a caveat. Three weeks was the right target for the MVP, she said. If the agent was tightly bounded. Low-value suppliers. Low-risk jurisdictions. Well-documented supplier categories, a deliberately small and measured sandbox. You can move fast when your blast radius is small. And Rebecca nodded slowly. So the discipline isn't the speed. The discipline is the scope. That turned out to be the sentence that unlocked the meeting.

Who manages the agent, and how will we know it's working?

MCDONAGH-SMITH: The last hour of the meeting was spent on two linked questions, and I want to take them in the order that the team did. The first was governance, specifically the architecture of human oversight. Olu laid out what she called the threshold map. For any supplier below a defined value and in a low-risk jurisdiction, the agent could run the full qualification and present the result to a human reviewer for a final sign-off. For any supplier above that value or in a flagged jurisdiction, the agent's role was reduced. It could prepare the analysis, but a human made the actual decisions. And for any supplier on a sanctions-related flag, the agent did nothing. It escalated immediately. "Human-in-the-loop is not a single setting," Olu said.

She remarked, “It’s a set of thresholds, and the thresholds are how we translate our risk appetite into code.”

Rebecca raised the second question: “How will we know it’s working?” This is where she pushed back on her own department’s measurement tradition. “The temptation,” she said, “was to compare the agent to the existing process on a single metric.” Probably cycle time. That comparison mattered but it wasn’t sufficient. The agent would change the shape of the work. Some things the agent did quickly would have knock-on effects further down the process that the old metric wouldn’t and couldn’t catch. Some of the value of the agent would show up not in costs saved but in capacity released. Procurement managers, newly freed to pursue suppliers that Boulton Industries had previously perhaps declined to evaluate.

She proposed a measurement approach the team hadn’t used before: continuous evaluation. Not a pre-deployment test that proved the system worked, followed by a production phase in which the system was assumed to keep working. Instead, a live dashboard in which the agent’s outputs, decisions, and drift were tracked continuously against a rolling baseline and reviewed monthly by a cross-functional group.

Vikram called this by the name the wider industry was beginning to use, agent operations or AgentOps, and said it was one of the few genuinely new disciplines the team would need to develop. He said the old model had been build, test, deploy. The new model is build, test, deploy, monitor, retrain, repeat continuously.

Priya brought in the question I’d been waiting for her to ask: “Who does that?” Who monitors it? Who notices when and if it drifts? Who decides when to intervene? That question, “Who manages the agent,” was the one that the team hadn’t yet answered. Rebecca said, “I’ve been thinking about this all week. I don’t think this is an engineering role. The engineering team will build the agents and keep it running. But the person who decides whether the agent’s decisions are still good, whether the agent is still doing what Boulton needs it to do, that has to be someone inside procurement. Someone who knows what a good supplier decision actually looks like. The engineering team isn’t equipped to make that judgment.”

Priya agreed and took the point further. She said, “This means we’re creating a new kind of role, part procurement specialist, part agent supervisor, and will need to create a lot of them over time as we deploy more of these systems. We will not be hiring for this; we will be growing for it.”

At this point, the CEO wrote a phrase on the whiteboard underneath the still visible, ambitious but unprepared line. She wrote, “Every agent needs a manager.” I watched that phrase land around the table. Maria nodded. Rebecca wrote it down. Olu said quietly, “That changes the organizational chart.”

The meeting closed with a decision. Boulton Industries would proceed with the supplier qualification agent. The scope would be deliberately bounded. Low-value suppliers, low-risk jurisdictions, three weeks to MVP, with Olu’s threshold map defining what the agent could and could not decide on its own. The measurement approach would be AgentOps from Day 1, with Rebecca’s team defining the dashboard. The agent would be managed day-to-day by a procurement specialist whom Priya and Rebecca would jointly identify,

with the role formally recognized as a new position rather than an add-on to an existing one.

The CEO then set the homework for the third Friday: “If the supplier qualification pilot teaches us what we expect it to teach us, if it works, what does it take to responsibly deploy 20 of these across the company within 18 months?” Bring your best thinking on that. We’ll meet next Friday at nine o’clock in the morning. The board recommendation has to be ready by the end of that day.

Key takeaways

MCDONAGH-SMITH: Now, let me offer a few observations. As the person who sat through that second meeting and watched the shape of the design emerge, what the Boulton team did in that meeting room was, in effect, to translate the abstract vocabulary of Week 1 into the specific architecture of a real agent. They picked a candidate not because it was the easiest but because it was the one that would force them to make the most decisions. They bounded the problem deliberately rather than chasing a universal solution. They worked through the five building blocks: perception, action, planning, memory, and safety as architectural decisions, not as features. They made a stack decision rather than a build or buy decision. They set a pace that was fast enough to create pressure, but slow enough to permit learning. And they acknowledged openly that the agent they were building would require a new kind of role to manage it. A role that Boulton Industries did not yet have and would need to grow into.

So if I were to compress what that second meeting surfaced into a short set of propositions, it would be these: An agentic system’s usefulness is a function of its boundaries, not its capabilities. The five building blocks of perception, action, planning, memory, and safety are architectural decisions each organization has to make deliberately, not features each organization can assume a vendor has chosen well. Build, buy, and stack are not equivalent. Each is a different bet on what you are trying to learn. The three-week MVP cadence is disciplined, not reckless, but only when the scope is held tight. Human-in-the-loop is not a single switch. It’s a set of thresholds, and the thresholds are how you translate your risk appetite into operational code.

Continuous evaluation, what the industry is calling AgentOps, is not a dashboard; it’s a discipline. And every agent in every organization will need to be managed, which is a role that a number of organizations haven’t yet thought to create. The team had designed their first agent on a whiteboard. They hadn’t yet deployed it. They hadn’t yet faced the harder question. The one the CEO had set for the following Friday: “What would it take to do this 20 times across the company responsibly at scale?” And that’s the question Episode 3 will follow them into.